

Investigating Compression and Size Biases in AI-Generated Image Detection

Kai-Hsuan Chan

University of Illinois Urbana-Champaign
Champaign, Illinois, USA
kc114@illinois.edu

Abstract

AI-generated images are becoming increasingly realistic, making automatic detection an important task for limiting misinformation and synthetic media misuse. However, recent work suggests that high benchmark performance may not always indicate that a detector has learned generation-specific artifacts. Instead, detectors may exploit dataset biases such as JPEG compression differences and image size distributions. This project studies these two biases in AI-generated image detection by reproducing and extending experiments inspired by *Fake or JPEG? Revealing Common Biases in Generated Image Detection Datasets*. Using subsets of GenImage, I evaluate detectors under controlled compression and size settings. The compression experiments compare detectors trained on raw images against detectors trained with generated images compressed to JPEG quality 96. The size experiments compare standard training against size-constrained training in which natural images are selected from a size interval similar to 512×512 generated images. I evaluate performance using accuracy, recall, and ROC-AUC. Results show that JPEG96-constrained training can substantially improve robustness under JPEG-compressed testing, especially for BigGAN and GLIDE settings. Size-constrained training produces more mixed results: it reduces performance on some 512×512 in-domain tests but improves recall on several cross-size test generators. Overall, the experiments show that AI-generated image detectors can rely on compression and size shortcuts, and that controlling these variables changes both robustness and cross-generator behavior.

Keywords

AI-generated image detection, dataset bias, JPEG compression, size bias, GenImage, robustness

1 Introduction

Recent advances in generative models have made it possible to create highly realistic synthetic images with little technical effort. These images appear in social media feeds, short-form video platforms, news discussions, and online advertisements. While image generation has many legitimate uses, it also creates risks: fake images can support misinformation, impersonation, political manipulation, or misleading evidence. For this reason, detecting whether an image is natural or AI-generated is an important computer vision problem.

A common approach is to train a binary classifier that predicts whether an image is real or fake. However, high performance on a benchmark does not necessarily mean that a detector has learned meaningful generative artifacts. If the real and fake images differ in simple properties such as file format, JPEG compression level, or image resolution, the detector may learn these easier cues instead

of learning features caused by the image generation process. This is especially concerning because real-world images are often resized, recompressed, uploaded to social media, or edited before detection.

This project focuses on two dataset biases in AI-generated image detection:

- (1) **JPEG compression bias:** natural images are often stored as JPEG files with lossy compression, while generated images are often saved as PNG files or otherwise less compressed.
- (2) **Image size bias:** images from a specific generator often have a fixed output resolution, while natural images have more varied dimensions.

The main research question is:

Do AI-generated image detectors learn compression and size shortcuts, and can controlled training reduce these biases?

The project is based primarily on *Fake or JPEG? Revealing Common Biases in Generated Image Detection Datasets* [1], which argues that AI-generated image detection datasets may contain unwanted compression and size biases. I reproduce a reduced version of this idea using GenImage [2], a large benchmark containing real ImageNet images and generated images from multiple image generators.

2 Motivation and Related Work

2.1 Why AI-Generated Image Detection Matters

Generative models have improved rapidly, especially with the rise of diffusion-based text-to-image systems. As generated images become more photorealistic, humans and automatic systems both face a harder discrimination problem. GenImage was proposed to address the lack of large-scale, diverse benchmarks for AI-generated image detection, especially for modern generators beyond early GAN-based systems [2].

A detector that only works under one dataset’s specific file properties is not reliable in practice. For example, if a model classifies uncompressed images as fake and JPEG-compressed images as real, then an attacker could simply recompress a generated image to evade detection. Similarly, if a model learns that 512×512 images are fake because a training generator always outputs that size, then performance may collapse when natural images have the same resolution or when generated images are resized.

2.2 Current State of the Art

Early work on synthetic image detection found that CNN-generated images can often be detected using standard convolutional classifiers. Wang et al. [6] showed that CNN-generated images are surprisingly easy to spot when training and testing conditions are

sufficiently aligned. Their work also demonstrated that augmentations such as JPEG compression and blur can improve generalization. However, later work suggests that such detectors may not generalize as well to newer generator families.

DIRE [7] proposed detecting diffusion-generated images by comparing an input image with its reconstruction through a diffusion model. This method is more directly targeted at diffusion models and moves beyond simply training a binary CNN classifier. However, reconstruction-based methods may be more expensive and may depend on the availability and suitability of a diffusion model used for reconstruction.

Ojha et al. [3] studied universal fake image detectors that generalize across generative models. They found that deep classifiers trained directly for real-vs-fake classification can become overly tuned to the artifacts of the fake generator seen during training. They proposed using feature spaces not explicitly trained for fake detection, such as CLIP features, with nearest-neighbor or linear probing methods.

The GenImage benchmark [2] introduced a large-scale dataset containing real ImageNet images and generated images from eight generators. It evaluates cross-generator image classification, where a detector trained on one generator is tested on others, and degraded image classification, where images are degraded through transformations such as JPEG compression, blur, and low resolution.

Most directly related to this project, Grommelt et al. [1] showed that GenImage and other AI-generated image detection datasets contain compression and size biases. They demonstrated that detectors can exploit these biases, and that controlling compression and size during training can improve robustness and change cross-generator performance.

3 Background

This project requires background knowledge in image classification, convolutional neural network detectors, ImageNet-style visual benchmarks, JPEG compression artifacts, and image resizing. These topics are necessary for understanding why a detector may learn compression or size shortcuts instead of learning true generation-specific artifacts.

The two main background papers used in this section are distinct from the three main motivation papers discussed above. The motivation section focuses on AI-generated image detection papers: *Fake or JPEG?* [1], GenImage [2], and universal fake image detection [3]. In contrast, this background section uses ImageNet [4] and ResNet [5] to explain the visual recognition and detector foundations needed to understand this project.

3.1 Binary Image Classification and ResNet Detectors

AI-generated image detection is commonly formulated as a binary image classification problem. Given an input image x , the detector predicts whether the image is real or AI-generated. A standard detector is a convolutional neural network trained with cross entropy or binary cross entropy loss.

A key background paper for this setup is He et al.’s ResNet paper [5]. ResNet introduced residual connections, which allow

very deep convolutional networks to be trained more effectively by learning residual mappings rather than direct transformations. This architecture became a standard backbone for image classification and many later computer vision tasks.

ResNet is important for this project because prior AI-generated image detection work often uses ResNet-50 as a baseline classifier. A ResNet-style detector has enough capacity to learn useful visual features, but it can also learn unintended shortcuts if those shortcuts are predictive during training. For example, if generated images are mostly uncompressed PNG files and real images are mostly JPEG-compressed, a ResNet detector may learn compression artifacts rather than generation artifacts. Therefore, understanding ResNet helps explain why strong benchmark performance does not automatically imply robust fake-image detection.

3.2 ImageNet and Class-Balanced Visual Benchmarks

Another important background paper is Deng et al.’s ImageNet paper [4]. ImageNet introduced a large-scale hierarchical image dataset with many object categories and became a foundation for modern visual recognition. It is widely used for image classification, pretraining, and benchmark construction.

ImageNet is important for this project because GenImage uses ImageNet real images and ImageNet class labels to construct a large-scale real/fake benchmark. The generated images are produced using ImageNet labels as prompts or class conditions, which helps align the semantic content of real and fake images. This is important because the detector should learn differences between real and generated images rather than simply learning that certain object categories are more likely to be fake.

However, ImageNet also introduces properties that matter for bias analysis. Real ImageNet images have diverse original sizes and are commonly stored with JPEG compression. These properties are not part of the semantic image label, but they can become correlated with the real class in an AI-generated image detection dataset. Therefore, understanding ImageNet as the real-image source helps explain why compression and size bias can appear in GenImage-style benchmarks.

3.3 JPEG Compression Bias

JPEG compression is a lossy transformation that changes image statistics by introducing compression artifacts. In many AI-generated image detection datasets, real images are collected from web-scale datasets and are often JPEG-compressed, while generated images are frequently saved as PNG files or with different compression properties. If the detector observes this pattern during training, it may learn that JPEG artifacts indicate real images and the absence of such artifacts indicates generated images.

This shortcut is problematic because it is not actually detecting whether the image was generated by an AI model. Instead, it detects a dataset artifact. If a generated image is recompressed as JPEG, the detector may incorrectly classify it as real. This motivates the JPEG-constrained training experiment in this project.

3.4 Image Size Bias

Image size can create a similar shortcut. Many image generators produce outputs at fixed resolutions, such as 512×512, while natural images from ImageNet have more varied dimensions. Even when all images are resized before being input to a neural network, the original size and aspect ratio can affect interpolation artifacts and visual statistics.

This motivates the size-constrained training experiment. By selecting natural images whose dimensions are close to the generated images, the training data reduces the correlation between image size and class label. The goal is to make the detector rely less on size-related artifacts and more on generation-specific features.

4 Problem Formulation

This project treats AI-generated image detection as a binary classification problem. Given an image x , a detector f_θ outputs a score or probability:

$$\hat{p} = f_\theta(x), \quad (1)$$

where $\hat{p}$ is interpreted as the probability that x is AI-generated. The predicted label is

$$\hat{y} = \begin{cases} 1, & \hat{p} \geq \tau, \\ 0, & \hat{p} < \tau, \end{cases} \quad (2)$$

where $y = 1$ denotes an AI-generated image, $y = 0$ denotes a real image, and τ is the classification threshold, usually set to 0.5.

A standard detector is trained with binary cross entropy:

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N [y_i \log f_\theta(x_i) + (1 - y_i) \log(1 - f_\theta(x_i))]. \quad (3)$$

This objective encourages correct real/fake classification, but it does not specify *which* image features the model should use. If compression level or image size is correlated with the label, the model may learn those shortcuts.

5 Key Technological Innovation

This project does not propose a new neural network architecture. Instead, the main focus is a controlled experimental and training protocol for testing whether an AI-generated image detector is relying on unwanted dataset biases. The key idea is to compare detector behavior under biased and constrained settings.

For compression bias, I compare:

- (1) **Raw training:** train detectors on the original GenImage-style data.
- (2) **JPEG-constrained training:** compress generated images to JPEG quality 96 during training so that generated images are closer to the natural image compression distribution.

For size bias, I compare:

- (1) **Standard size training:** train with natural images of any size.
- (2) **Size-constrained training:** select natural images whose dimensions fall within a controlled interval, such as [450, 550], to better match 512×512 generated images.

The purpose of these constraints is not to make the task artificially easier. Instead, the constraints remove simple label-correlated shortcuts so that the detector is pushed toward learning features more closely related to synthetic generation.

6 Dataset

6.1 GenImage

The main dataset is GenImage [2]. GenImage contains 2,681,167 images in total, including 1,331,167 real images and 1,350,000 fake images. The real images come from ImageNet, and the fake images are generated using eight models:

- Midjourney
- Stable Diffusion V1.4
- Stable Diffusion V1.5
- ADM
- GLIDE
- Wukong
- VQDM
- BigGAN

The dataset is class-balanced over 1000 ImageNet labels. This is useful because the detector should not solve the task by learning class imbalance. Each GenImage subset pairs real images with images generated by a particular generator. This structure supports cross-generator evaluation: train on one generator subset and test on another.

6.2 Subset Used in This Project

Due to the limited project scope, I used a reduced subset of GenImage instead of the full dataset. For the compression-bias experiments, I focused on the following generators:

- GLIDE
- ADM
- BigGAN

These generators provide a manageable reduced experiment while still including both diffusion-based models and a GAN-based model.

For the size-bias experiments, I focused on generators with similar output sizes:

- Wukong
- Stable Diffusion V1.4
- Stable Diffusion V1.5

These generators produce 512×512 images, making them appropriate for testing whether matching the natural-image size distribution reduces size bias. I also evaluate these size-constrained detectors on smaller-output generators, including BigGAN, ADM, and GLIDE, to observe whether the size constraint changes cross-generator behavior.

6.3 Labels and Metadata

Each datum has a binary label:

$$y = \begin{cases} 0, & \text{real image,} \\ 1, & \text{AI-generated image.} \end{cases} \quad (4)$$

Additional useful metadata includes:

- generator name,
- original image size,
- file format,
- compression setting, if produced by the experiment,
- ImageNet class label.

6.4 Dataset License and Access

GenImage is publicly described in the GenImage paper [2]. According to the dataset license terms, unless specifically labeled otherwise, the datasets are provided under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International Public License, or CC BY-NC-SA 4.0, with additional dataset-specific terms. The license terms state that the dataset may be used only for non-commercial purposes, including academic research, teaching, and scientific publications. The terms also prohibit commercial use of the dataset or derivative works, including selling the data or using it for commercial gain.

This license is appropriate for this project because the experiments are conducted for academic research and course-report purposes. No commercial use is involved.

7 Evaluation Metrics

I use three evaluation metrics: accuracy, recall, and ROC-AUC.

7.1 Accuracy

Accuracy measures the fraction of correctly classified examples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (5)$$

where TP is the number of generated images correctly classified as generated, TN is the number of real images correctly classified as real, FP is the number of real images incorrectly classified as generated, and FN is the number of generated images incorrectly classified as real.

Accuracy is useful because the real and fake classes are approximately balanced in GenImage. However, accuracy alone can hide whether a detector is failing mainly on real images or fake images.

7.2 Recall

For the generated-image class, recall is:

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (6)$$

Recall is especially important for this project because compression bias often causes generated images to be misclassified as real after recompression. In that case, fake-image recall decreases even if precision remains high.

7.3 ROC-AUC

ROC-AUC measures how well the detector ranks generated images above real images across all possible thresholds. Let $s(x)$ be the detector’s generated-image score. ROC-AUC can be interpreted as:

$$\text{AUC} = P(s(x^+) > s(x^-)), \quad (7)$$

where x^+ is a randomly chosen generated image and x^- is a randomly chosen real image. Unlike accuracy, ROC-AUC is threshold-independent.

8 Baselines

8.1 Baseline 1: Raw Detector

The first baseline is a detector trained on raw GenImage-style data. This baseline represents the standard training condition where the detector may observe existing compression and size correlations between real and generated images.

8.2 Baseline 2: JPEG-Constrained Detector

The second baseline follows the compression-mitigation idea from *Fake or JPEG?* [1]. Generated images are compressed to JPEG quality 96 during training. This reduces the compression gap between natural and generated images.

8.3 Baseline 3: Size-Constrained Detector

The third baseline is used for the size-bias experiment. Natural images are filtered to include only images whose dimensions fall within a range such as [450, 550]. This makes the natural-image size distribution more similar to the 512×512 generated images used in the size experiment.

8.4 Published Baselines

Published baselines discussed in this report include:

- CNN-based detection from Wang et al. [6].
- DIRE for diffusion-generated image detection [7].
- CLIP-feature-based universal fake detection from Ojha et al. [3].
- GenImage benchmark results [2].
- Bias-mitigated GenImage experiments from Grommelt et al. [1].

9 Experiments

9.1 Experiment 1: Compression Bias

The first experiment tests whether detector performance changes when test images are recompressed. I train detectors under two settings:

- (1) Raw training.
- (2) JPEG96-constrained training.

Then I evaluate the trained detectors on test images under different compression settings:

- raw/original,
- JPEG quality 96,
- JPEG quality 95,
- JPEG quality 90.

The matrices in this experiment use the training generator as the row and the testing generator as the column. The main comparison is the difference between JPEG96-constrained training and raw training. Positive values mean that JPEG96-constrained training performed better than raw training.

9.2 Experiment 2: Size Bias

The second experiment tests whether detectors rely on image size. I compare:

- (1) Standard training using natural images of any size.
- (2) Size-constrained training using natural images with dimensions in the interval [450, 550].

The generated training subsets are selected from 512×512 generators: Wukong, Stable Diffusion V1.4, and Stable Diffusion V1.5. The evaluation includes these 512×512 generators and smaller-output generators such as BigGAN, ADM, and GLIDE.

9.3 Implementation

All testing code for the image compression and size-bias experiments was written by me. The implementation includes code for applying JPEG compression, filtering images by size interval, training detectors, evaluating checkpoints, computing accuracy, recall, and ROC-AUC, and producing the heatmap plots used in this report.

The project repository is available at:

<https://github.com/shayne773/robust-ai-image-detection>.
git

10 Numerical Results

10.1 Compression-Bias Results

The compression results show that JPEG96-constrained training improves robustness when test images are compressed. The strongest improvements occur under JPEG90, the most compressed test condition.

For accuracy, the largest improvement is observed when training on BigGAN and testing on BigGAN under JPEG90: JPEG96-constrained training improves accuracy by 47.66 percentage points over raw training. There are also notable improvements for BigGAN training tested on GLIDE under JPEG90, with a 14.39 point improvement, and for GLIDE training tested on GLIDE under JPEG90, with a 14.18 point improvement.

Recall shows an even stronger pattern. Under JPEG90, BigGAN trained and tested on BigGAN improves by 95.40 percentage points when using JPEG96-constrained training instead of raw training. BigGAN trained on BigGAN and tested on GLIDE improves by 28.77 points, while GLIDE trained and tested on GLIDE improves by 28.38 points. This suggests that raw-trained detectors often fail to identify compressed generated images as fake.

ROC-AUC also improves under compression. Under JPEG90, BigGAN trained and tested on BigGAN improves by 27.96 points. GLIDE trained on GLIDE and tested on BigGAN improves by 11.60 points, and BigGAN trained on BigGAN and tested on ADM improves by 14.11 points. This means that JPEG96-constrained training does not only improve thresholded accuracy and recall; it can also improve the detector’s ranking of real and fake images.

The raw test condition is more mixed. For example, under raw testing, GLIDE training evaluated on BigGAN loses 10.47 points in accuracy and 20.98 points in recall after JPEG96-constrained training. This indicates a tradeoff: JPEG96-constrained training improves robustness to recompression, but it may reduce performance in some raw cross-generator settings where the raw detector benefits from the original dataset’s compression differences.

Table 1: Selected compression-bias improvements from JPEG96-constrained training over raw training. Values are percentage-point differences.

Test Setting	Train/Test	Acc.	Recall	AUC
JPEG90	BigGAN/BigGAN	+47.66	+95.40	+27.96
JPEG90	BigGAN/GLIDE	+14.39	+28.77	+7.09
JPEG90	GLIDE/GLIDE	+14.18	+28.38	+1.72
JPEG95	BigGAN/BigGAN	+40.62	+81.35	+6.62
JPEG96	BigGAN/BigGAN	+36.13	+72.37	+3.69
Raw	GLIDE/BigGAN	-10.47	-20.98	+1.14

10.2 Feature-Space Analysis of JPEG Compression

To better understand why JPEG-constrained training changes detector behavior, I also visualize the detector feature space using PCA. Figure 4 compares GLIDE raw images and GLIDE JPEG90-degraded images under two different detectors: a raw-trained detector and a JPEG90-trained detector. Each point represents an image embedding projected onto the first two principal components. The colors and markers distinguish raw real images, raw fake images, JPEG90-degraded real images, and JPEG90-degraded fake images.

In the raw-trained detector, the fake-image cluster shifts noticeably when the fake images are degraded with JPEG90 compression. This supports the quantitative result that raw-trained detectors are sensitive to compression changes. The degraded fake points move closer to regions that are not as clearly separated from real-image structure, which can explain why recall drops under JPEG compression.

In contrast, the JPEG90-trained detector produces a feature space where the real and fake groups remain more consistently separated even after JPEG90 degradation. The real images occupy a long curved region, while the fake images are concentrated in a different region on the left side of the plot. This suggests that JPEG-constrained training changes the internal representation: instead of treating compression artifacts as the main signal, the detector learns a feature space that is more stable when fake images are recompressed.

This visualization provides qualitative support for the compression-bias results. The heatmaps show that JPEG-constrained training improves accuracy, recall, and ROC-AUC under compressed testing, while the PCA plot shows how the learned embedding space changes when compression is included during training.

10.3 Size-Bias Results

The size-constrained experiment shows a more complex pattern than the compression experiment. For the 512×512 generator group, size-constrained training keeps high absolute performance. For example, the size-constrained detector achieves 93.49 accuracy for SDv5 trained and tested on SDv5, 94.46 accuracy for SDv4 trained and tested on SDv4, and 91.56 accuracy for Wukong trained and tested on Wukong. ROC-AUC is also high within the 512×512 group, often above 96.

However, the difference from standard training shows that size-constrained training does not always improve accuracy. Within

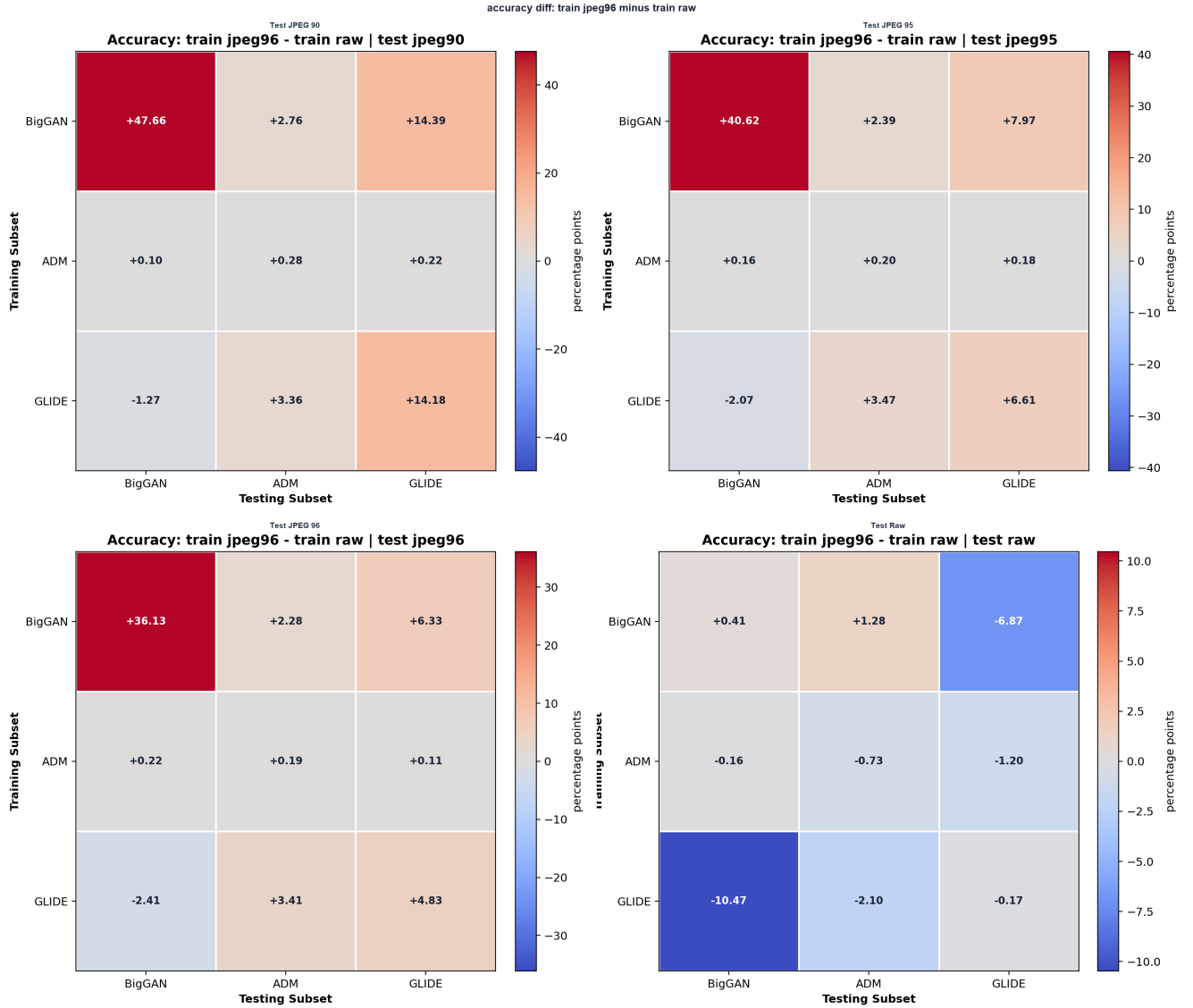


Figure 1: Accuracy difference between JPEG96-constrained training and raw training across test compression settings. Positive values mean that JPEG96-constrained training improves accuracy over raw training. The largest improvement appears for BigGAN trained and tested under JPEG90, where accuracy improves by 47.66 percentage points.

the 512×512 group, accuracy often decreases by around 4 to 8 percentage points. For example, Wukong trained and tested on SDv5 decreases by 7.60 points, and Wukong trained and tested on SDv4 decreases by 7.34 points. This suggests that part of the standard detector’s performance may have come from size-related cues that are reduced by the size constraint.

At the same time, size-constrained training improves recall on several cross-size evaluations. SDv5 trained and tested on ADM improves recall by 22.03 points. SDv4 trained and tested on GLIDE improves recall by 22.33 points. SDv4 trained and tested on ADM improves recall by 21.85 points. These recall improvements suggest

that size-constrained training can make the detector less conservative about identifying generated images from some smaller-output generators.

The ROC-AUC results are mixed. Size-constrained training improves ROC-AUC for SDv4 tested on ADM by 13.06 points and SDv5 tested on ADM by 11.11 points. However, it decreases ROC-AUC for Wukong tested on BigGAN by 17.07 points and SDv5 tested on GLIDE by 18.80 points. This suggests that size constraint alone does not universally improve cross-generator generalization, but it significantly changes what the detector learns.

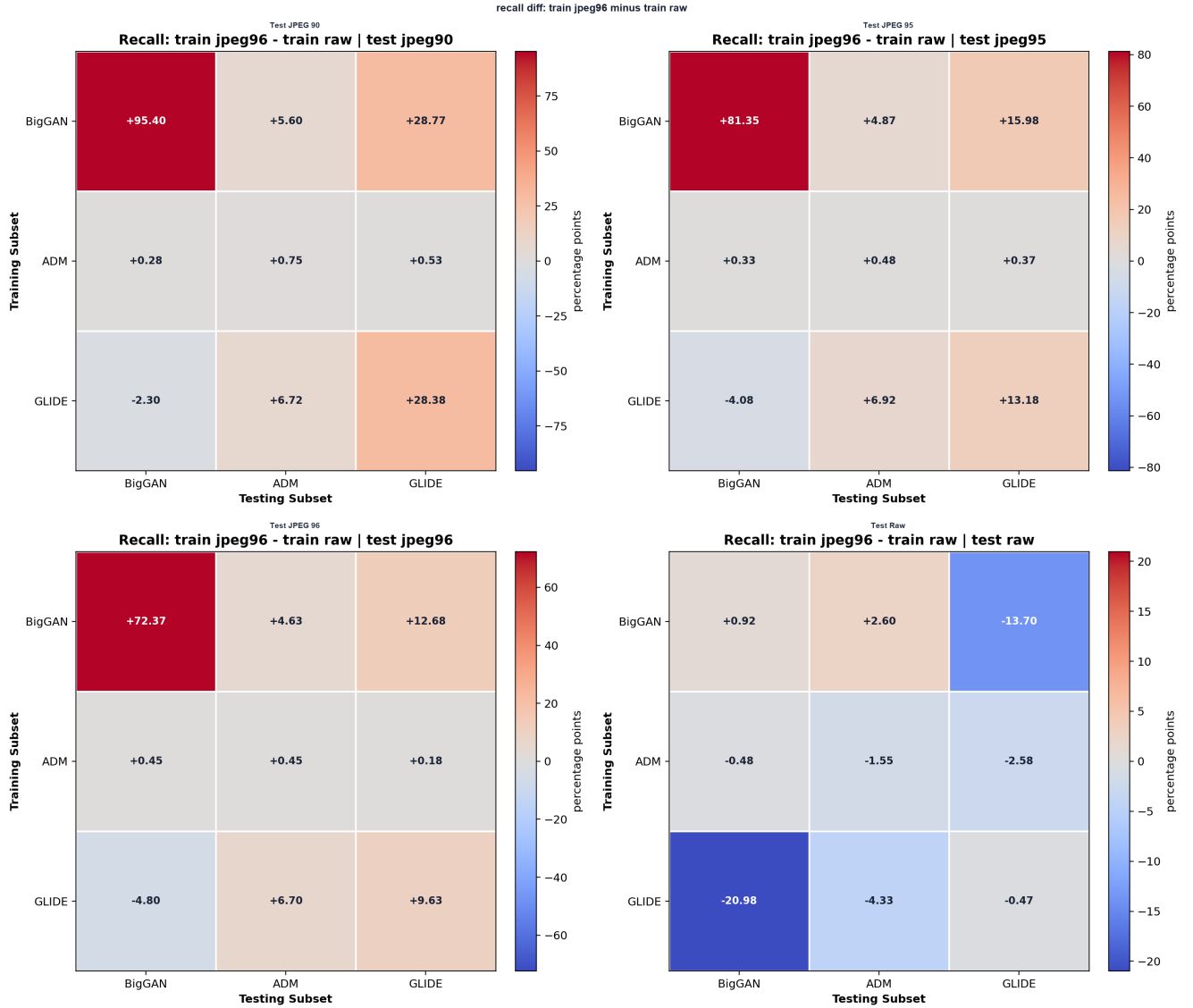


Figure 2: Recall difference between JPEG96-constrained training and raw training across test compression settings. Recall shows the strongest evidence of compression bias. For BigGAN trained and tested under JPEG90, JPEG96-constrained training improves recall by 95.40 percentage points.

11 Ablation Studies

11.1 Ablation 1: Compression Level

The compression ablation modifies the JPEG quality applied at test time. The independent variable is the JPEG quality factor, and the dependent variables are accuracy, recall, and ROC-AUC. This ablation directly tests whether a detector’s prediction changes when generated images are recompressed.

The results show that raw-trained detectors are vulnerable to compression changes. The most extreme case is BigGAN under JPEG90 testing, where JPEG96-constrained training improves recall by 95.40 percentage points. This indicates that the raw-trained

detector was not simply learning generator-specific features; it was also relying on compression-related differences between the real and generated classes.

The JPEG96-constrained detector is more robust because the training process reduces the compression gap between real and generated images. Since generated images are compressed during training, the model cannot rely as easily on the absence of JPEG artifacts as a fake-image cue.

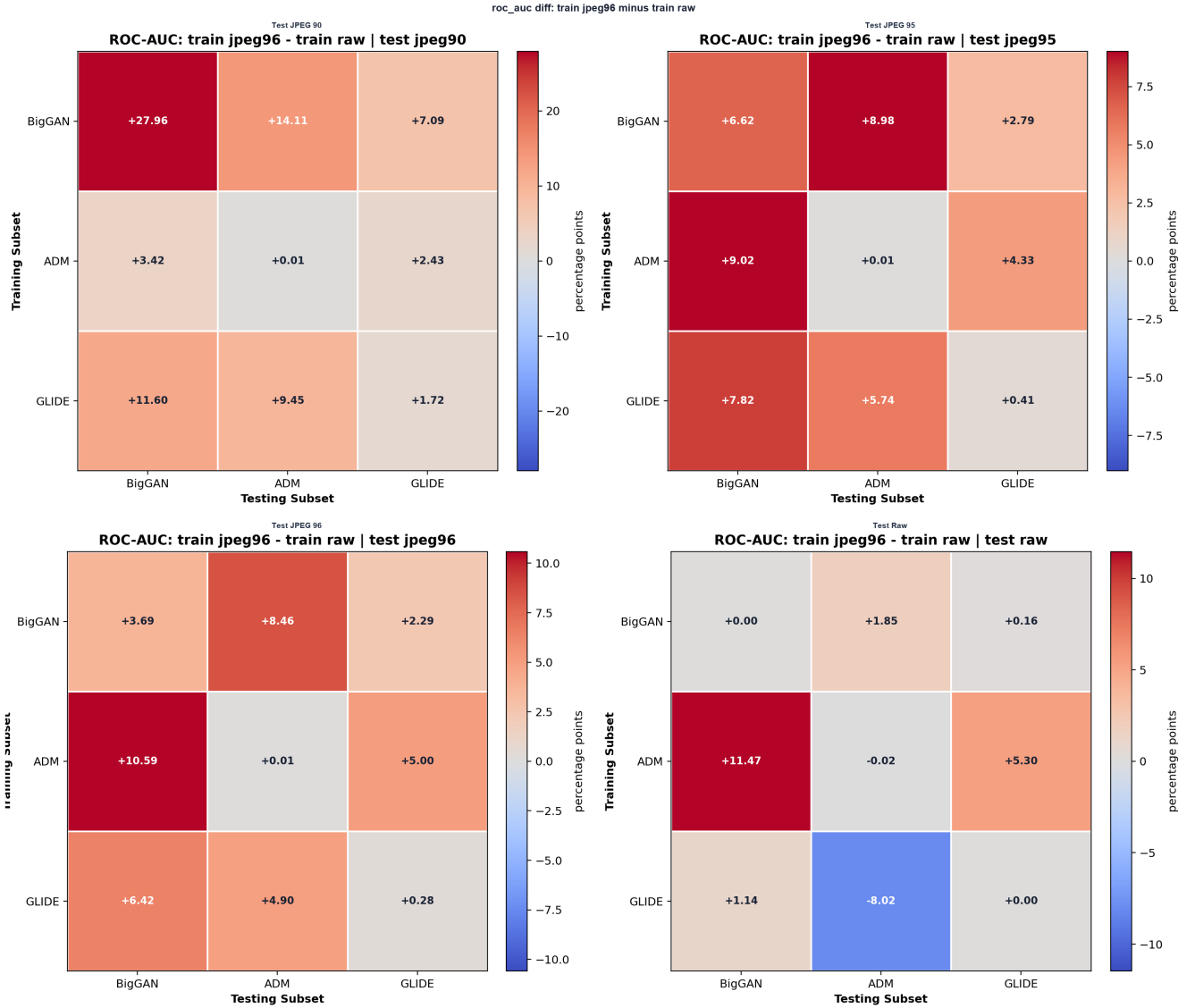


Figure 3: ROC-AUC difference between JPEG96-constrained training and raw training across test compression settings. JPEG96-constrained training generally improves ranking quality under compressed testing, especially under JPEG90.

11.2 Ablation 2: Natural Image Size Filtering

The size ablation modifies the natural-image selection rule. In the standard condition, natural images of any size are allowed. In the constrained condition, natural images are selected only when their dimensions fall within the desired interval.

The results show that size-constrained training affects different metrics differently. Accuracy within the 512×512 group decreases compared with standard training, suggesting that the standard detector may have benefited from size-related shortcuts. However, recall improves on several cross-size tests, especially for ADM and GLIDE. This means that removing the size shortcut can make the detector more willing to classify generated images from other

generators as fake, even though this does not always translate into higher ROC-AUC or accuracy.

12 Discussion

The central scientific generalization from this project is that AI-generated image detectors can learn dataset shortcuts when real and fake images differ in non-causal properties. Compression and size are not the same as generation artifacts. However, if they are correlated with the real/fake label during training, a standard classifier has no reason to ignore them.

The compression experiment provides the clearest evidence of shortcut learning. When the raw-trained detector is evaluated on

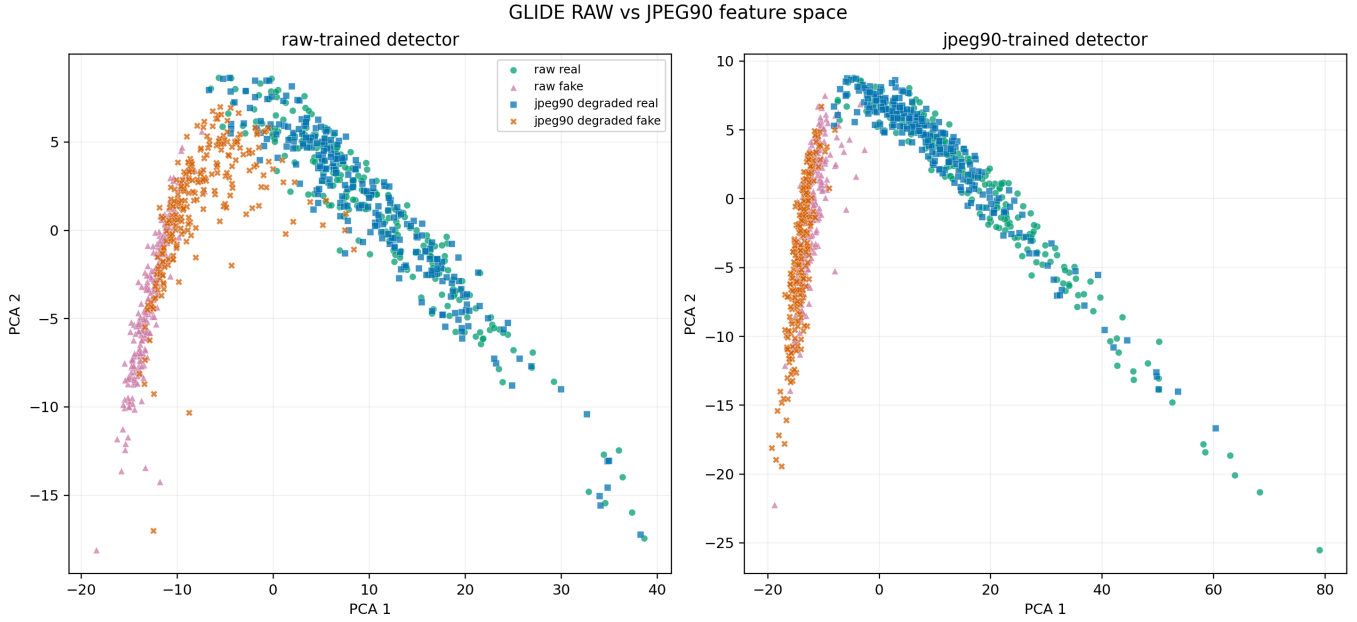


Figure 4: PCA visualization of GLIDE raw and JPEG90-degraded image embeddings. The left plot shows the feature space of a raw-trained detector, while the right plot shows the feature space of a JPEG90-trained detector. The raw-trained detector shows more overlap between raw fake images and JPEG90-degraded fake images, while the JPEG90-trained detector separates the real-image cluster and fake-image cluster more clearly under compression.

Table 2: Selected size-constrained results. Difference values compare size-constrained training against standard training in percentage points.

Train/Test	Metric	Constrained	Difference
SDv5/SDv5	Accuracy	93.49	-5.29
SDv4/SDv4	Accuracy	94.46	-4.51
Wukong/Wukong	Accuracy	91.56	-6.63
SDv5/ADM	Recall	23.28	+22.03
SDv4/GLIDE	Recall	27.27	+22.33
SDv4/ADM	ROC-AUC	69.34	+13.06
Wukong/BigGAN	ROC-AUC	41.96	-17.07
SDv5/GLIDE	ROC-AUC	56.90	-18.80

compressed test data, performance can collapse. The strongest example is BigGAN under JPEG90 testing, where JPEG96-constrained training improves recall by 95.40 percentage points. This suggests that the raw detector was strongly affected by compression differences. In practical settings, this is dangerous because uploaded images are often recompressed by websites and social media platforms.

The feature-space visualization further supports this conclusion. The PCA plot shows that JPEG degradation changes where fake images appear in the embedding space of the raw-trained detector. When the detector is trained with JPEG compression, the separation between real and fake images becomes more stable under the same JPEG90 degradation. This indicates that compression-constrained

training affects not only the final metric values, but also the internal representation learned by the detector.

The size experiment is more nuanced. Size-constrained training does not uniformly improve all metrics. Instead, it reveals that standard training may have benefited from original size differences, especially within the 512×512 generator group. After reducing the size shortcut, some in-domain accuracy decreases. However, recall improves for several cross-size generator tests, especially for ADM and GLIDE. This supports the idea that controlling size changes the detector’s decision behavior and can reduce reliance on one dataset-specific cue.

These findings also help explain why cross-generator evaluation is difficult. A detector trained on one generator may learn the generator’s size, compression, or preprocessing pattern. When tested on another generator, those shortcuts may no longer apply. Therefore, better evaluation should control for dataset-level artifacts, not only report aggregate accuracy.

13 Conclusion

This project investigates JPEG compression bias and image size bias in AI-generated image detection. Using GenImage subsets, I compare raw training against compression-constrained and size-constrained training. The compression experiment shows that JPEG96-constrained training can substantially improve robustness under recompressed testing. The largest observed improvement is a 95.40 percentage-point recall gain for BigGAN trained and tested under JPEG90.

The PCA feature-space analysis provides additional qualitative evidence for this conclusion. Under the raw-trained detector, JPEG90

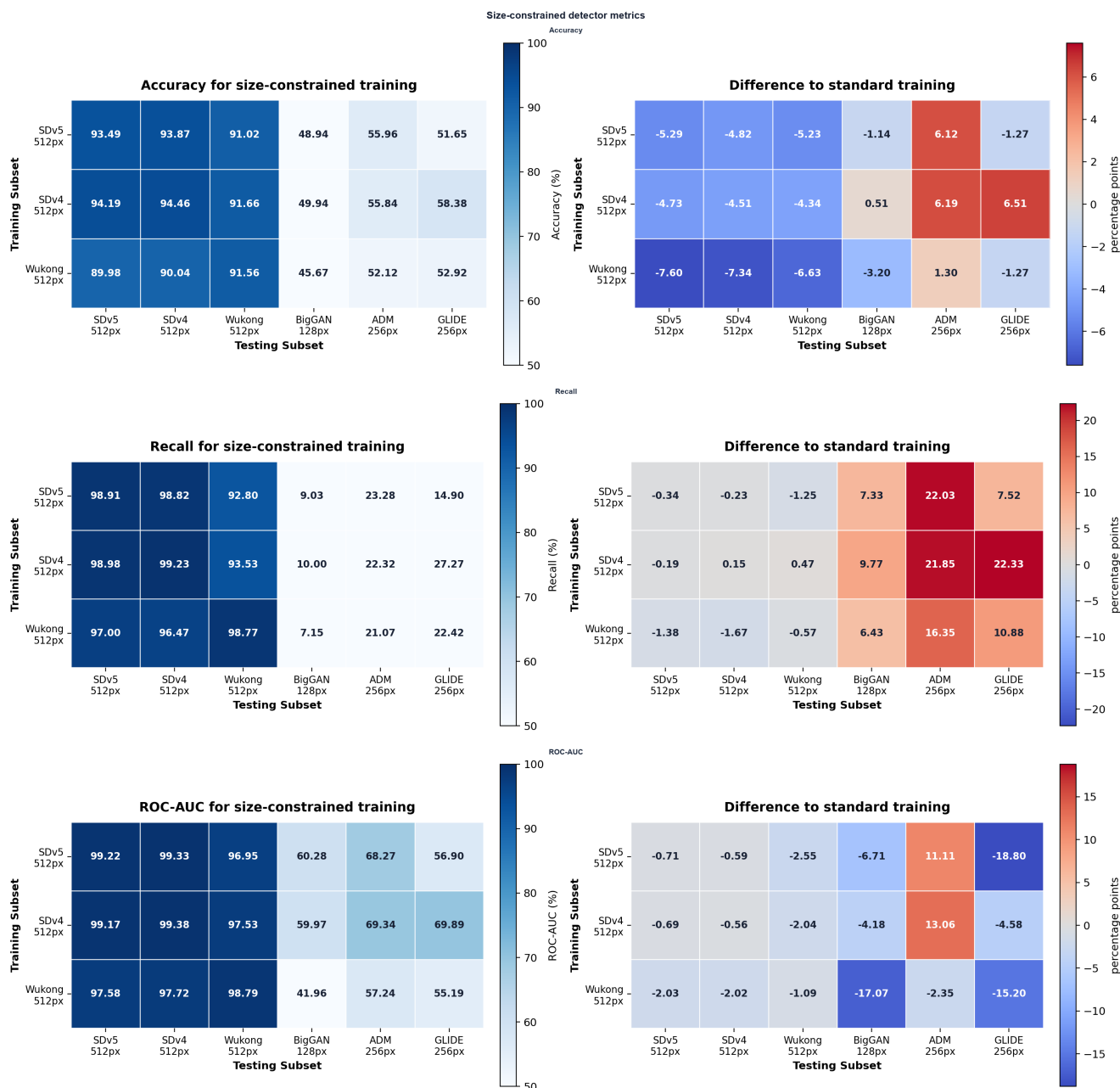


Figure 5: Accuracy, recall, and ROC-AUC for size-constrained training and the difference from standard training. The left column shows the absolute performance of the size-constrained detector, while the right column shows the change relative to standard training. Size-constrained training maintains high performance among 512×512 generators, but it reduces some in-domain accuracy while improving recall on several cross-size tests.

degradation changes the embedding distribution of fake images, while the JPEG-trained detector produces a more stable separation between real and fake groups under compression.

The size experiment shows that size-constrained training has mixed effects. It reduces some accuracy on 512×512 generator tests,

but it improves recall on several cross-size tests, including gains above 20 percentage points for SDv5 or SDv4 tested on ADM or GLIDE. This suggests that size constraints reduce certain shortcuts but do not automatically solve cross-generator generalization.

The main reason for these differences is that constrained training changes which features are predictive. When compression and size are correlated with the label, a detector can exploit them. When those variables are controlled, the detector is forced to rely less on spurious dataset artifacts and more on other visual evidence.

14 Statement of Contributions

All testing code for the image compression and size-bias experiments was written by Kai-Hsuan Chan. This includes the code for JPEG compression evaluation, size filtering, model evaluation, metric computation, embedding-space visualization, and plotting. The project uses the GenImage dataset and is inspired by the experimental design of *Fake or JPEG? Revealing Common Biases in Generated Image Detection Datasets*. The implementation repository is available at:

<https://github.com/shayne773/robust-ai-image-detection>.
git

The GenImage dataset is used under the CC BY-NC-SA 4.0 license with additional dataset terms. The dataset is used only for non-commercial academic research and course-report purposes.

Initial report draft: Kai-Hsuan Chan.

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